



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 16/07/2026

EXPERIMENT 5 -- SELECT with various clause – WHERE, pattern matching

AIM:

To retrieve and display records from the **Student_CGPA** table using the **SELECT** statement with **WHERE** clause and **LIKE** operator for filtering data based on conditions and pattern matching in MySQL Workbench.

PROCEDURE:

1. Open **MySQL Workbench** and connect to the MySQL server.
2. Create a new database and select it using the USE command.
3. Create the **student_cgpa** table with the required columns and constraints.
4. Insert sample records into the **student_cgpa** table using the INSERT statement.
5. Execute **SELECT** statements to display data from the table.
6. Use the **WHERE** clause to retrieve records based on specific conditions such as department, city, CGPA, and number of failed subjects.
7. Use the **LIKE** operator with wildcard characters (%) and (_) to perform pattern matching on student names and city names.
8. Execute each query in MySQL Workbench and verify the output displayed in the Result Grid.
9. Analyze the retrieved records to understand how the **WHERE** clause and **LIKE** operator filter data.
10. Conclude the experiment after successfully obtaining the required outputs.

PROGRAM:

1. CREATE TABLE

```
CREATE TABLE student_cgpa (  
    reg_no VARCHAR(15) PRIMARY KEY,  
    student_name VARCHAR(50) NOT NULL,  
    department VARCHAR(20),  
    no_of_subs_pass INT,  
    no_of_subs_fail INT,  
    cgpa DECIMAL(3,2)  
);
```

Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)  
VALUES  
(192411123, 'Deepthi', 'CSE', 22, 2, 8.71),  
(192512121, 'Varun', 'IT', 27, 1, 9.23),  
(192312117, 'Aadhya', 'ECE', 25, 3, 8.97),  
(192411227, 'Dimple', 'CSE', 31, 0, 9.37),
```

```
( '192820896','Mohan','EEE',18,9,6.48),
('192411122','Yogitha','CSE',31,1,9.01),
( '192516871','Sadhya','IT',25,5,7.22),
('192111101','Rahul','MECH',30,12,5.14),
('192411010','Alekhyia','CSE',35,2,9.01),
('192511135','Rahul','CIVIL',37,0,9.72);
```

```
SELECT * FROM student_cgpa;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhyia	ECE	25	3	8.97
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48

#	Time	Action	Message	Duration / Fetch
1	22:41:47	CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...	0 row(s) affected	0.031 sec
2	22:44:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	22:46:18	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

WHERE Clause

1. Display Registration Number, Student Name and CGPA of all students.

```
SELECT reg_no, student_name, cgpa
FROM student_cgpa;
```

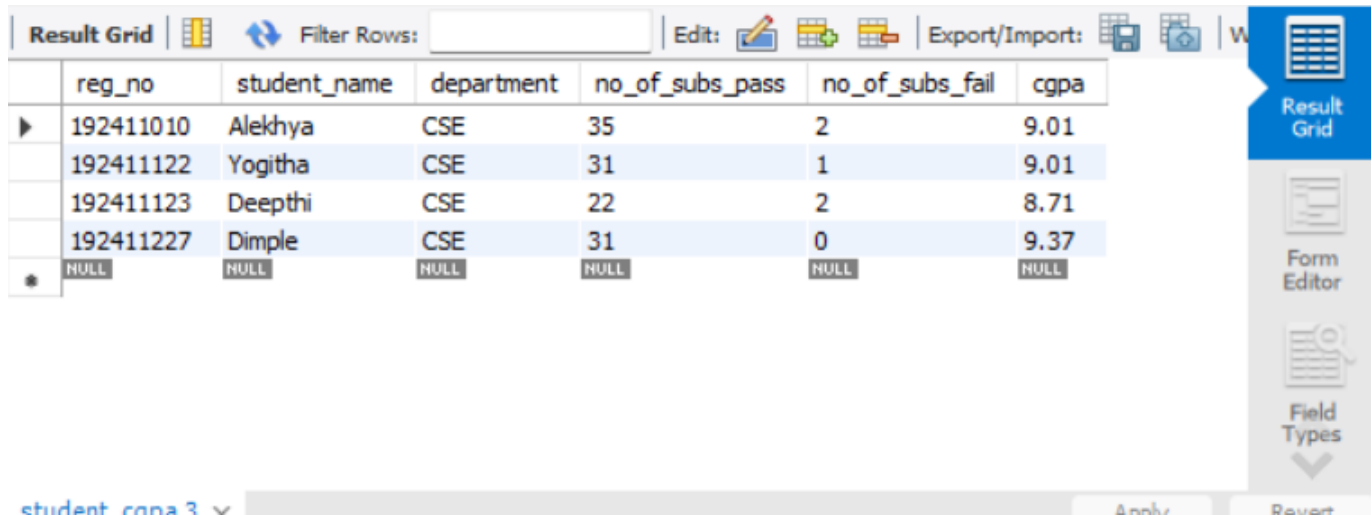
OUTPUT:

reg_no	student_name	cgpa
192312117	Aadhyia	8.97
192411010	Alekhya	9.01
192411122	Yogitha	9.01
192411123	Deepthi	8.71
192411227	Dimple	9.37
192511135	Rahul	9.72
192512121	Varun	9.23
192516871	Sadhya	7.22
192820896	Mohan	6.48
NULL	NULL	NULL

2. Display all students from the CSE department.

```
SELECT *  
FROM student_cgpa  
WHERE department = 'CSE';
```

OUTPUT:



The screenshot shows a database application interface. At the top, there is a toolbar with buttons for 'Result Grid', 'Filter Rows', 'Edit', and 'Export/Import'. Below the toolbar is a table with the following data:

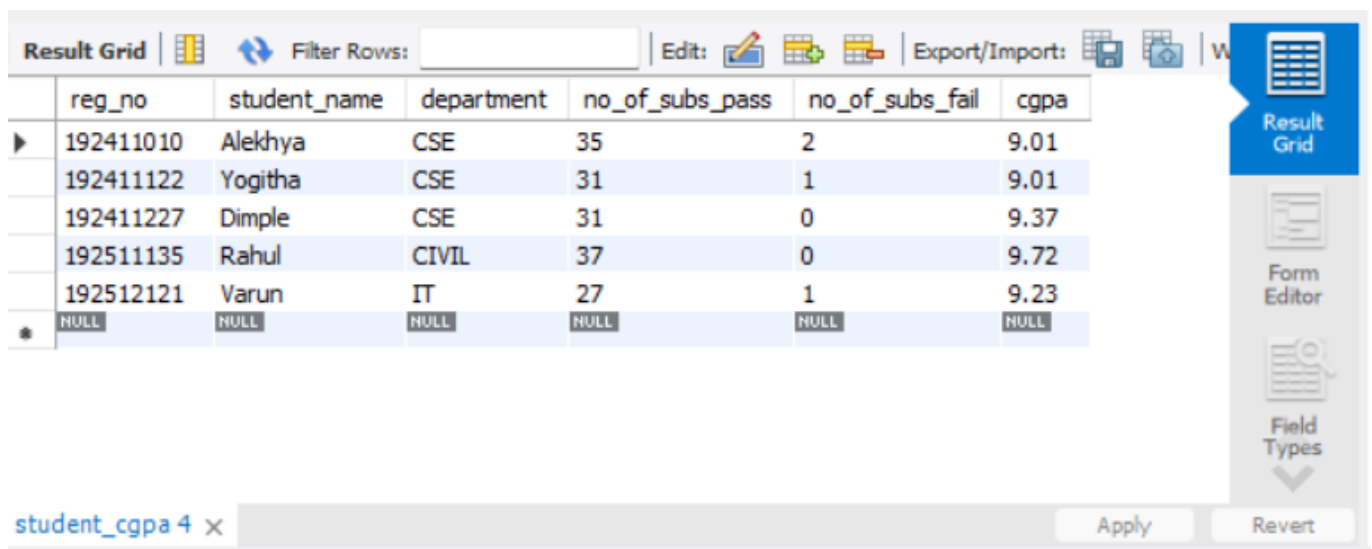
	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
▶	192411010	Alekhyia	CSE	35	2	9.01
	192411122	Yogitha	CSE	31	1	9.01
	192411123	Deepthi	CSE	22	2	8.71
	192411227	Dimple	CSE	31	0	9.37
*	NULL	NULL	NULL	NULL	NULL	NULL

On the right side of the interface, there is a vertical toolbar with buttons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom, there is a status bar with the text 'student_cgpa 3' and buttons for 'Apply' and 'Revert'.

3. Display students whose CGPA is greater than 9.00.

```
SELECT *  
FROM student_cgpa  
WHERE cgpa > 9.00;
```

OUTPUT:



The screenshot shows a database application interface. At the top, there is a toolbar with buttons for 'Result Grid', 'Filter Rows', 'Edit', and 'Export/Import'. Below the toolbar is a table with the following data:

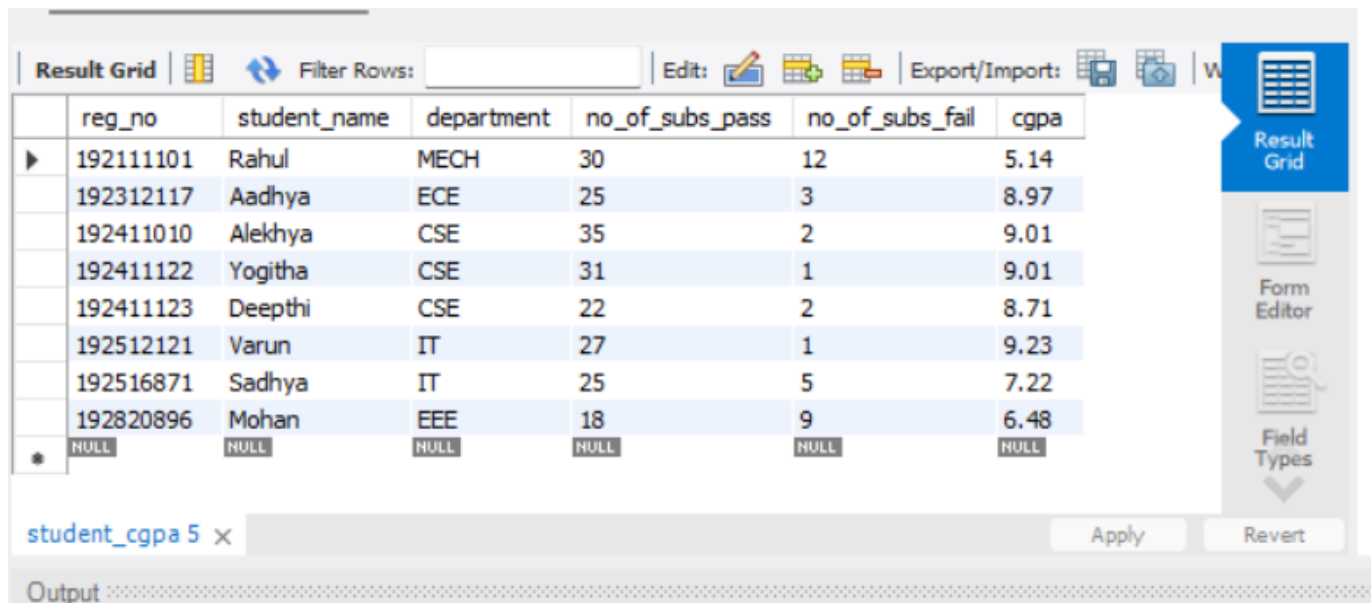
	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
▶	192411010	Alekhyia	CSE	35	2	9.01
	192411122	Yogitha	CSE	31	1	9.01
	192411227	Dimple	CSE	31	0	9.37
	192511135	Rahul	CIVIL	37	0	9.72
	192512121	Varun	IT	27	1	9.23
*	NULL	NULL	NULL	NULL	NULL	NULL

On the right side of the interface, there is a vertical toolbar with buttons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom, there is a status bar with the text 'student_cgpa 4' and buttons for 'Apply' and 'Revert'.

4. Display students who have failed in one or more subjects.

```
SELECT *  
FROM student_cgpa  
WHERE no_of_subs_fail > 0;
```

OUTPUT:



	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
▶	192111101	Rahul	MECH	30	12	5.14
	192312117	Aadhya	ECE	25	3	8.97
	192411010	Alekhyia	CSE	35	2	9.01
	192411122	Yogitha	CSE	31	1	9.01
	192411123	Deepthi	CSE	22	2	8.71
	192512121	Varun	IT	27	1	9.23
	192516871	Sadhya	IT	25	5	7.22
	192820896	Mohan	EEE	18	9	6.48
*	NULL	NULL	NULL	NULL	NULL	NULL

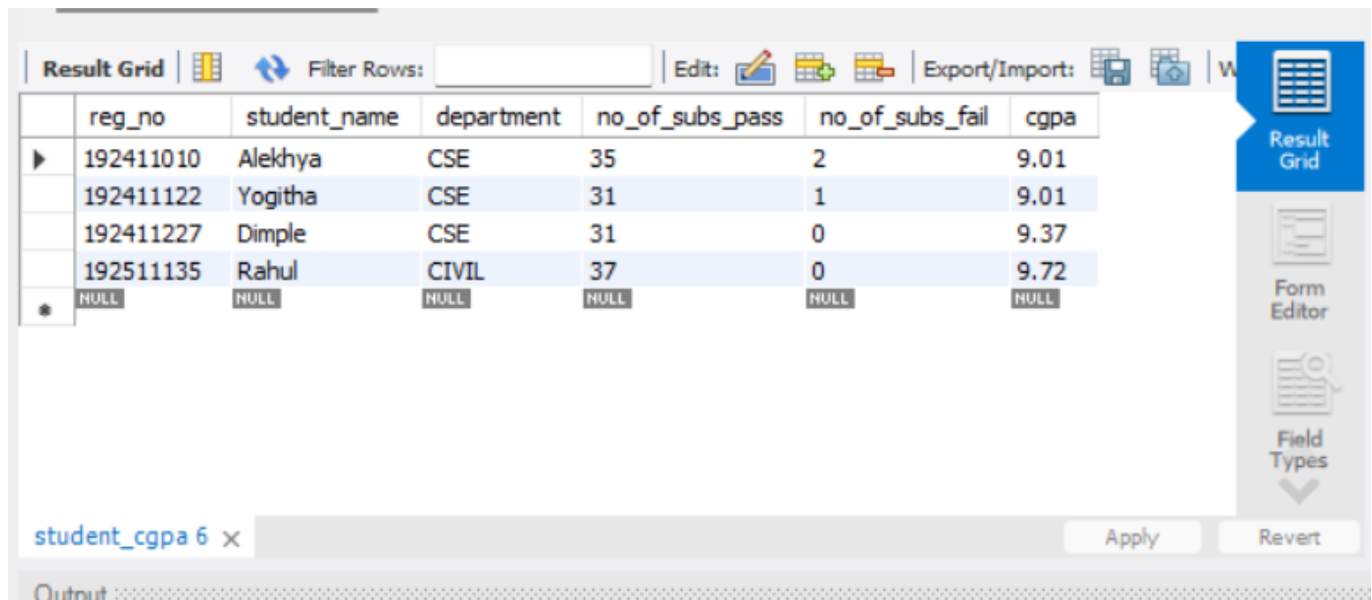
student_cgpa 5 x Apply Revert

Output

5. Display students who have passed more than 30 subjects.

```
SELECT *  
FROM student_cgpa  
WHERE no_of_subs_pass > 30;
```

OUTPUT:



	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
▶	192411010	Alekhyia	CSE	35	2	9.01
	192411122	Yogitha	CSE	31	1	9.01
	192411227	Dimple	CSE	31	0	9.37
	192511135	Rahul	CIVIL	37	0	9.72
*	NULL	NULL	NULL	NULL	NULL	NULL

student_cgpa 6 x Apply Revert

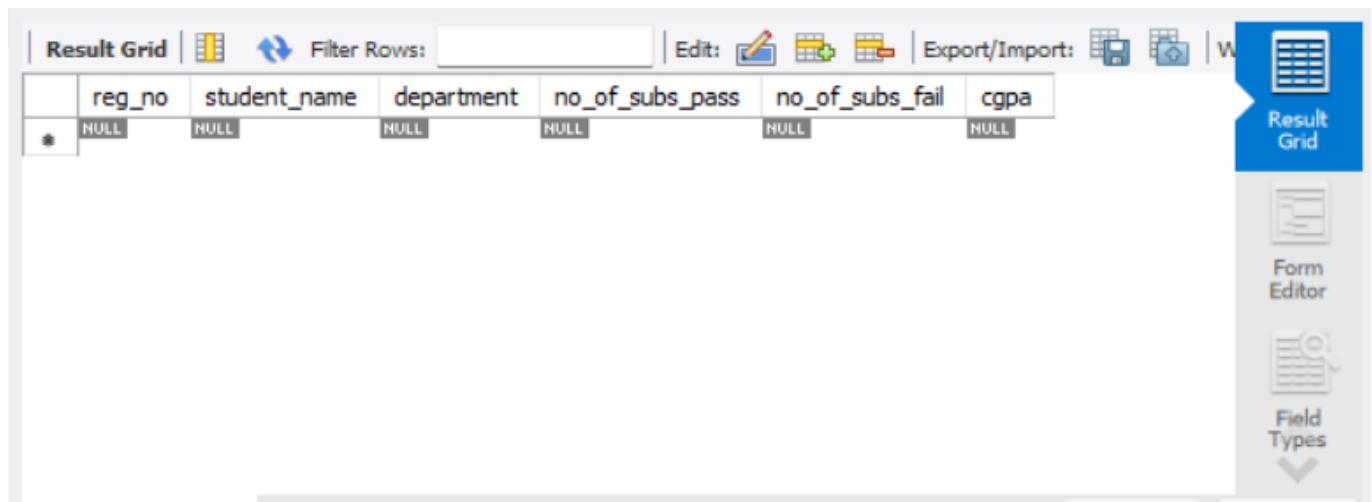
Output

LIKE Clause

6. Display students whose names end with "sh".

```
SELECT *  
FROM student_cgpa  
WHERE student_name LIKE '%sh';
```

OUTPUT:



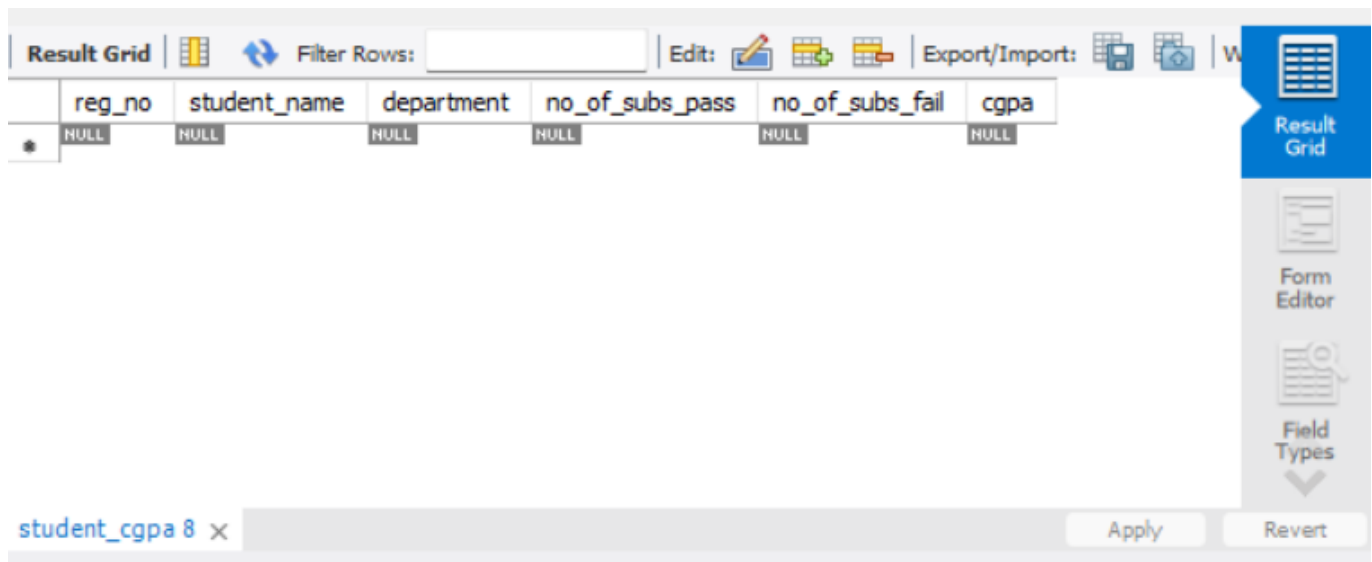
The screenshot shows a database application window. At the top, there is a toolbar with icons for 'Result Grid', 'Filter Rows', 'Edit', and 'Export/Import'. Below the toolbar is a table with the following columns: 'reg_no', 'student_name', 'department', 'no_of_subs_pass', 'no_of_subs_fail', and 'cgpa'. The first row of the table contains the value 'NULL' for each of these columns. On the right side of the window, there is a vertical sidebar with buttons for 'Result Grid', 'Form Editor', and 'Field Types'.

	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
*	NULL	NULL	NULL	NULL	NULL	NULL

7. Display students whose names contain "sh".

```
SELECT *  
FROM student_cgpa  
WHERE student_name LIKE '%sh%';
```

OUTPUT:



The screenshot shows the same database application window as before. The table now displays one row with the following values: 'reg_no' is NULL, 'student_name' is NULL, 'department' is NULL, 'no_of_subs_pass' is NULL, 'no_of_subs_fail' is NULL, and 'cgpa' is NULL. The bottom of the window shows a status bar with the text 'student_cgpa 8 x' and two buttons: 'Apply' and 'Revert'.

	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
*	NULL	NULL	NULL	NULL	NULL	NULL

8. Display students whose department contains "CS".

```
SELECT *  
FROM student_cgpa  
WHERE department LIKE '%CS%';
```

OUTPUT:

Result Grid						
Filter Rows:						
	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
▶	192411010	Alekhyia	CSE	35	2	9.01
	192411122	Yogitha	CSE	31	1	9.01
	192411123	Deepthi	CSE	22	2	8.71
	192411227	Dimple	CSE	31	0	9.37
*	NULL	NULL	NULL	NULL	NULL	NULL

student_cgpa 9 x Apply Revert

9. Display students whose names do not contain six letters.

```
SELECT *
FROM student_cgpa
WHERE student_name NOT LIKE '_____';
```

OUTPUT:

Result Grid						
Filter Rows:						
	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
▶	192111101	Rahul	MECH	30	12	5.14
	192411010	Alekhyia	CSE	35	2	9.01
	192411122	Yogitha	CSE	31	1	9.01
	192411123	Deepthi	CSE	22	2	8.71
	192511135	Rahul	CIVIL	37	0	9.72
	192512121	Varun	IT	27	1	9.23
	192820896	Mohan	EEE	18	9	6.48
*	NULL	NULL	NULL	NULL	NULL	NULL

student_cgpa 10 x Apply Revert

10. Display students whose names contain "am".

```
SELECT *
FROM student_cgpa
WHERE student_name LIKE '%am%';
```

OUTPUT:

Result Grid						
Filter Rows:						
Edit:   						
Export/Import:  						
	reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
*	NULL	NULL	NULL	NULL	NULL	NULL

RESULT:

Thus, the records from the student_cgpa table were successfully retrieved using the SELECT statement with WHERE clause and LIKE pattern matching in MySQL Workbench.